

Lydia Mesa

MEng student, Télécom Paris (Institut Polytechnique de Paris) | Product & Strategy · Design · Software & AI

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Native French & Japanese (JLPT N1) · English C1 (Cambridge) · German C1 (DSD II)

Education

Télécom Paris, Institut Polytechnique de Paris

Palaiseau, France

MEng / Diplôme d'Ingénieur - 5-year program conferring grade de Master

2024 - 2028

- Dual major: **Data Science & AI + 3D Graphics & Interaction (HCI)** · expected graduation **June 2028**
- **Undergraduate cycle completed July 2026 (BSc equivalent)**, 240 ECTS validated · GPA **3.58 / 4.0** (final year 3.73) · #2 engineering school in France (QS 2026)
- Coursework: machine learning, computer vision, algorithms & data structures, databases, information visualisation, HCI

Lycée Michelet - *Classes Préparatoires (CPGE)*, filière PSI

Vanves, France

Two-year intensive program; national competitive entrance exam for the French engineering schools

2022 - 2024

Experience

ArianeGroup - IT Strategy & Software Engineering Intern

Bordeaux, France

Enterprise digital transformation: collaborative software landscape for European space launchers

Sep 2026 - Feb 2027

- Designing and fielding the **current-state technology analysis** behind a company-wide **re-architecture of the collaborative software stack** (8,700 employees, 5,000+ external contributors) - **200+ stakeholder interviews and surveys** establishing where stated preferences diverge from real needs
- Segmenting findings by team, role and workflow, then evaluating candidate solutions **cross-functionally** against business objectives and adoption constraints to build the **future-state vision and roadmap** toward 2029

EY - Strategy & Technology Consulting Intern

Paris-La Défense, France

Rotational programme: EY-Parthenon (Strategy), Consulting, Assurance & Financial Services

Jun - Jul 2025

- Scoped an open brief with no fixed deliverable (how teams adapt to new technology) down to **enterprise AI adoption**, and ran **12 stakeholder interviews** mapping tools, workflows, unmet needs and readiness
- **Synthesised findings into recommendations** on skills and tooling a non-technical audience could act on; **presented to EY managers**

Paris 2024 Olympic Games - Assistant to the President, Japanese Olympic Committee

Paris, France

VVIP operations & trilingual liaison (JP/FR/EN) between two institutional cultures

Jul - Aug 2024

Selected Projects - Product, Design & Engineering

Recette - Social Photo-Editing App

React Native, Expo, Supabase/Postgres, Skia (GLSL)

Solo product, design and build - a social app where every edit is a copyable recipe, reapplied in one tap

2026 - Present

- Designed **locked layers** to protect original-creator attribution, trading remixability for a credit model that holds; wrote the edit pipeline as **on-device Skia/GLSL shaders** rather than server-side rendering to keep the social loop latency-free
- Built a full **gamification economy** (points for posting and for being saved, a store to spend them in), then **chose not to ship it** once modelling showed it would compound early users' advantage and lock newcomers out
- Ran a **34-user closed beta** (~21% week-one return) and used the cohort data to reprioritise the roadmap

Voice-Driven VR Architecture Studio

Unity, Meta Quest, Gemini API · team of 3

Interaction & interface design lead - prototyping tool for architects who don't use 3D software

2026

- Replaced VR menu hierarchies with **natural-language voice control**, removing the controller fluency barrier for non-gamer users; designed a **keyword-summary confirmation step** so users see what the system understood before committing
- Found how people actually phrase requests through **7 design iterations with 9 users**, and deliberately **bounded the intent space** rather than accepting free-form commands - trading breadth for reliability

Review Sentiment Platform for a Social-Sciences Research Team

RoBERTa/BERT, Django, Leaflet.js

Scoped, built and handed over to a research group with no technical staff

Jan - Jun 2025

- Researchers wanted to measure how public perception of places shifts after external shocks, but had only unstructured review text - **fine-tuned a RoBERTa classifier** to label ~3,000 free-text reviews at **86% accuracy**
- Aggregated to location-level sentiment on an interactive map, letting the team quantify how **COVID lockdowns and transit strikes** shifted sentiment over time; handed over as a tool they could run without me

Assistive Ball-Tracking for Blind Football Spectators

Python, OpenCV, YOLOv9, Arduino

National research project - graded 20/20, top 3% nationally

2023 - 2024

- Conveyed live ball position through **touch instead of sight**; made latency the binding constraint and accepted bounded error - **180 ms inference**, ±6m localisation, enough to signal "left wing, near the box" on a 12cm haptic board
- Simulated **60 camera placements** to find the real coverage/accuracy trade-off rather than assuming an angle; scoped out full player tracking as cost without user value

Leadership

Comète (audiovisual club), Télécom Paris - Secretary General

2025 - 2026

Student media club publishing 50,000+ event photos a year - prototyped a facial-recognition photo-search tool, then **halted its launch** after a **GDPR** and biometric-data review

Télespoir (humanitarian club), Télécom Paris - Communications & Corporate Relations Lead

2024 - 2025

Raised **EUR 10,000** in corporate sponsorships for an annual health and education mission to Togo - cold outreach and sponsor pitches

Student Board, Télécom Paris - Travel & Audiovisuals Lead

2024 - 2026

Delivered a **90-person trip** on a fixed **EUR 20,000** budget with a team of 5, against a date that could not move

Skills

- **Product & strategy:** current-/future-state analysis, stakeholder research at scale, segmentation, prioritisation, roadmapping
- **Design & research:** interaction design (HCI), prototyping, usability testing, survey design · Figma, Illustrator, InDesign
- **Engineering:** Python, C++, Java, SQL, JavaScript/HTML/CSS, PyTorch, OpenCV, React Native, OpenGL/GLSL, Three.js, Unity, Git